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(20524) Roll No.

BCA-II Sem.

18006

B.C.A. Examination, May-2024

C-Programming

(BCA-202)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt **all** sections as per instructions.

Section-A

Note : Attempt **all 5** questions. Each question carries **3** marks. Very short answer is required not exceeding **75** words.

1. What are header files and their uses?
2. What is the difference between macro and functions?

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3. What is a structure? How it is different from Union?
4. What is static memory allocation and dynamic memory allocation?
5. What is the difference between getc(), getchar(), getch() and getche()?

Section-B

Note : Attempt all **2** questions out of the following **3** questions. Each question carries **7.5** marks. Short answer is required not exceeding **200** words.

6. Write a C program that reads 20 integer numbers from keywords and sort them in ascending order.
7. Write a C program to check input string is palindrome or not without using any string library function.

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8. Write a C program to add two complex numbers by passing structure to a function.

Section-C

Note : Attempt any **3** questions out of the following **5** questions. Each question carries **15** marks. Answer is required in details. https://www.ccsustudy.com

9. What do you mean by pointer? Write C program to access elements in a 2 dimensional array using pointer.

10. How string is different an array? Write a C program convert a lower case string to upper case string without using string manipulation function.

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11. Write a program in C to count the number of words and characters in an input file test 1.txt.

12. Write a program in C for the multiplication of two matrices of size 4×4.

13. Create a structure named 'Book' to store book details like title, author, and price. Write a C program to input details for three books, find the most expensive and the lowest priced books, and display their information.

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Printed Pages : 4

(20623)

Roll No.....

BCA-II Sem.

18006

B.C.A. Examination, June-2023

C-PROGRAMMING

(BCA-202)

Time : 3 Hours] *[Maximum Marks : 75*

Note : Attempt ***all*** the sections as per instructions.

Section-A

(Very Short Answer Type Questions)

Note : Attempt *all five* questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words.

1. Write various data type supported in C with example. 3

2. What are headers files ? Why they are important ? 3

3. Write limitations of switch case. 3

4. Differentiate between scope and lifetime of a variable. 3

5. Define preprocessor and its usage in programming. 3

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Section-B

(Short Answer Type Questions)

Note : Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words. 2×7½=15

6. Write short notes on following :

(i) Enumerated Data Type

(ii) String

(iii) Macro Expansion

(iv) File Inclusion

7. Write a program in C to print following pattern :

```
2  3  4  5  6  7
3  4  5  6  7
4  5  6  7
5  6  7
6  7
7
```

8. What are different file opening modes ? Write a program in C that reads a series of integer numbers from a file named INPUT and write all odd numbers to a file to be called ODD and all even numbers to a file to be called EVEN.

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Section-C

(Long Answer Type Questions)

Note : Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 3×15=45

9. (a) Write a program to calculate sum of the following series upto first 100 terms :

$\int = 1^4 + 3^4 + 5^4 + 7^4$ ----- 100 Terms.

(b) Given an array of 20 integers. Write a program in C to search for a given integer in that array.

10. (a) Write a program in C to multiply two matrics of N×N dimension.

(b) Declare a structure which contains the following members and write a program in C to list all students who scored more than 75 marks. Roll No., Name, Father's Name, Age, City, Marks

11. (a) Explain dynamic memory concept with proper example.

(b) Explain different type of modes and I/O function in file handling.

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12. (a) Define recursive function. Write a program to find factorial of a number with recursive function.

(b) Write a program to check whether a given number is Armstrong number or not.

13. (a) Write a program to copy a string to another string using standard library function. Use dynamic memory allocation to accept string.

(b) Differentiate between call by value and call by reference.

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(20622) Roll No.

BCA-II Sem.

18006

B.C.A. Examination, June-2022

C Programming

(BCA-202)

Time : Three Hours] *[Maximum Marks : 75*

Section-A

Note : *Attempt all* **five** questions. Each *question carries* **3** marks. Very short answer is required not exceeding 75 words.

1. Explain the need for array variables .
2. Distinguish between automatic and static variables.
3. What is meant by array of structures.
4. What is a pointer? How is a pointer initialized?
5. Describe the use and limitations of the function getc.

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Section-B

Note : Attempt any **two** questions. out of the following **three** questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words.

6. What is a data structure? Why is an array called a data structure? Write a program to read a matrix of size mxn and print its transpose.
7. Describe the three logical bitwise operators what is the purpose of each? What types of operands are required by each of the logical bitwise operators?
8. What is the relationship between an array name and a pointer? How is an array name interpreted when it appears as an argument to a function? How can a function return a pointer to its calling routine?

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Section-C

Note : Attempt any **three** questions. out of the following **five** questions. Each question carries **15** marks. Answer is required in detail.

9. (a) Character string in C are automatically terminated by the null character. Explain how this feature helps in string manipulations.

   (b) Main is a user defined function. How does is differ from other uses-defined functions?

10. (a) Compare the working of the function strcat and strncat. Write a program, which read your name from the key board and outputs a list of ASCII codes, which represent your name.

    (b) What are the rules that govern the passing of arrays to function? Use recursive function calls to evaluate

$$f(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots\ldots$$

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11. (a) Explain the meaning and purpose of the following:

    (i) Struct keyword

    (ii) Tyedef keyword

    (iii) Size of operator

    (b) Write a function that receive a sorted array of integers and an integer value, and inserts the value in its correct place.

12. (a) Define auto and register variables in the content of 'C'. What is the basic difference between these two variables?

    (b) Give examples of using feof and ferror in a program.

13. (a) What do you know about bitwise operator? Explain about some bit wise operators by providing the examples for each?

    (b) Define a macro that receives an array and the number of elements in the array as arguments. Write a program using this macro to print out the elements of an array.

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(20519) Roll No. .....

Total Questions : **13**] [Printed Pages : **4**

18006

B.C.A. IInd Semester Examination, May-2019

C-PROGRAMMING

(BCA-202)

Time : 3 Hrs.] *[M.M. : 75*

Note :– Attempt all the Sections as per instructions.

Section–A

(Very Short Answer Type Questions)

Note :– Attempt all *five* questions. Each question carries 3 marks. Very short answer is required not exceeding **75** words.

1. What are *three* dimensional arrays ? How can you initialize them ?
2. How a union is different from a structure ?
3. What do you mean by a dangling pointer ?

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4. What are bit fields ?
5. What are the use of standard functions fopen( ) and feof( ) ?

Section–B

(Short Answer Type Questions)

Note :– Attempt any *two* questions out of the following three questions. Each question carries 7.5 marks. Short answer is required not exceeding **200** words.

6. Write a program that will count the number occurrences of a specified character in a given line of text.
7. Write a program to pre-multiply a matrix by its transpose.
8. Design a structure named student to store data about a student which contains following data element :

Date Item	Type	Length
Roll No.	int	–
Name	char	20
College	char	40
Score	float	–

Assume that there are not more than 100 students. Write a program to input the data about students, and output the stored data according to the merit of the students.

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Section–C

(Long Answer Type Questions)

Note :– Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail.

9. (i) How is a multidimensional array defined in terms of an array pointer ? What does each pointer represent ? How does this definition differ from a pointer to a collection of contiguous arrays of lower dimensionality ?

(ii) What is meant by dynamic memory allocation ? What library function is used to allocate memory dynamically ? How is the size of the memory block specified ? What kind of information is returned by the library function ?

10. (i) Write short notes on the following :

(a) strlen( )

(b) strcpy( )

(c) strcat( )

(d) strcmp( )

(ii) What are the important points to be considered when implementing bit-fields in structures ?

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11. (i) Describe *two* different approaches to undating a data file. Which approach is better and why ? For what kinds of applications are unformatted data files well suited ?

(ii) Write a function using pointers to add two matrices and to return the resultant matrix to the calling function.

12. (i) What is a masking operation ? What is the purpose of each operand ? Which operand is the mask, and how is it chosen ?

(ii) Write macro definitions with arguments for calculation of simple Interest and Amount. Store these macro definitions in a file called 'Interest.n'. Include this file in your program, and use the macro definitions for calculating simple interest and amount.

13. (i) What are the difference between Union and Structure ?

(ii) Write short notes on the following :

(a) fewind( )

(b) fseek( )

(c) fgetc( )

(d) fscanf( )

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(20518) Roll No. .....

BCA-II Sem.

18006

B. C. A. Examination, May 2018

C Programming

(BCA-202)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 3×5=15

1. Differentiate between string and character array. 3

2. What is generic pointer ? How can it be converted to a specific type of pointer ? 3

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3. What is the output ? 3

```
#include < stdio.n>
  int func (int);
  main( )
{
  int a = 2;
  printf ("%", func(a));
  return 0;
}
int func(int a)
{
  if (a > 1)
  return func (- - a) + 10;
  else
  return 0;
}
```

4. Explain the difference between malloc ( ) and calloc( ) function. 3

5. Explain the importance of the # define preprocessor directive. 3

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three questions. Each question carries 7½ marks. Short answer is required not exceeding 200 words.

7½×2=15

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6. Write a program to sort an array. 7½

7. Write a C program to find the reverse of each word of a string (how are you : output : woh era uoy). 7½

8. Differentiate between rewind( ) and fseek( ). Can fseek( ) work as an alternative to rewind( ). If yes, why ? 7½

Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. (a) Why are arrays needed ? Write a program to calculate the number of duplicate entries in an array. 7½

   (b) With the help of an example, explain how pointers can be used to dynamically allocate space for two-dimensional array. 7½

10. (a) What do you understand by EOF ? Write a program to read a text file, convert all the lower case characters into upper case. 7½

    (b) What is string ? Explain any five library functions of string. 7½

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11. (a) Create a Structure BANK to maintain the records of a bank customers. It has the following fields CUST-ID, NAME, ACC-TYPE, BALANCE.

    (i) A new record is added when a customer open an account

    (ii) A existing record is updated when user deposits or withdraw an amount.

    Create Menu-Driven Program. 7½

    (b) What do you understand by pointers ? Write a program to count the number of characters, words and lines in the text using pointer. 7½

12. (a) What is macro ? Explain the difference between object macro and function macro with example. 5

    (b) Write a program to swap two numbers using pointer with structure. 10

13. (a) Explain the use of bitwise operators in programming with suitable example. 7½

    (b) Write a program that reads a binary file that stores employees records and prints on the screen the number of records that are stored in the file. 7½

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(20517) Roll No. .....

BCA-II Sem.

18006

B.C.A. Examination, May 2017

C Programming

(BCA-202)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from **all** sections as per instructions.

Section-A

Note : Attempt **all** the **five** questions. Each question carries **3** marks. Very short answer is required not exceeding 75 words.

3×5=15

1. Explain the difference between array and structure. 3

2. Differentiate between *(arr+i) and (arr+i). 3

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3. Write a short note on nested structures. 3

4. Explain command line arguments in C using example. 3

5. How pointers are implemented with function with example. 3

Section-B

Note : Attempt any two questions:

6. Write the output: 7½

```
#include <stdio. h>
main ()
{
    int arr [ ] = {1, 2, 3, 4, 5};
    int i=1, j=2;
    Printf ("% d"  *(arr+1+i));
    printf ("% d"  *(arr+*(arr+1)));
    printf ("% d",  *(arr+j));
    printf ("% d",  *(arr+i)+*(arr+j)));
}
```


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7. Write a program to interchange second element with last element of an array. 7½

8. Write short note on: 7½

(a) fopen ( )

(b) fclose ( )

(c) fgetc ( )

(d) fprintf ( )

Section-C

Note : Attempt any **three** questions.

9. (a) What is string? Write a program to find concatenate of two string using pointers without Library function. 7½+7½=15

(b) Write a program using pointers to search a value from an array.

10. (a) Write a short note on conditional directives. 5+10=15

(b) In a class there are 5 students. Each student is supposed to appear in 3 tests. Write a program using 2-D array to print.

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(i) the marks obtained by each student in different subjects.

(ii) total marks and average obtained by each student.

11. (a) Explain different types of files in file handling and what are different modes to open a file? 10+5=15

(b) Difference between structure and union. Explain how members of a union are accessed using a program code.

12. (a) What is file? Write C program to copy the contents of one file into another file. 7½+7½=15

(b) Explain different bitwise shift operators use in C Programming.

13. (a) What is pointer? Explain pointer to pointer with example? Can we subtract two pointer variables. 7½+7½=15

(b) Write a program to explain the use of structure with function.

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V

(20516) Roll No. .....

BCA-II Sem.

18006

B. C. A. Examination, May 2016

C Programming

(BCA-202)

(New)

Time : Three Hours] *[Maximum Marks : 75*

Note : Attempt questions from all Sections as per instructions.

Section-A

(Very Short Answer Questions)

Attempt all the *five* questions. Each question carries 3 marks. Very short answer is required not exceeding 75 words. 3×5=15

1. What are two-dimensional arrays ? How can you initialize them ?

2. How does a structure differ from an array ?

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3. Distinguish between (*m) [5] and *m[5].

4. How can you declare and initialize a string ?

5. Explain the meaning of the following :
   (i) Tag
   (ii) Size of.

Section-B

(Short Answer Questions)

Attempt any *two* questions out of the following three question Each question carries 7½ marks. Short answer is required not exceeding 200 words. 7½×2=15

6. What is a dynamic array ? How is it created ? Give a typical example of use of a dynamic array.

7. Write a program to illustrate the comparison of structure variables.

8. What is a pointer ? How can it be initialized ? Write a program using pointer to read in an array of integers and print its elements in reverse order.

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Section-C

(Detailed Answer Questions)

Attempt any *three* questions out of the following five questions. Each question carries 15 marks. Answer is required in detail. 15×3=45

9. (a) Give the main advantage of storing data as a file. Describe various ways in which data files can be categorized in 'C'. Illustrate by using examples.

   (b) Enumerate the difference between functions and parameterized macros.

   Why do we recommend the use of parentheses for formal arguments used in a macro definition?

10. (a) Describe the two bitwise shift operators. What requirement must be operands satisfy ? What is the purpose of each operand ?

    (b) Describe two different approaches to updating a data file. Which approach is better and why ? For what kinds of applications are unformatted data files well suited ?

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11. (a) Write a program to illustrate the use of structure pointers.

    (b) What are jumping statements ? Explain the difference between break and continue statements.

12. (a) Describe the limitation of using getchar and scanf functions for reading strings. Write a program which will read a text and count all occurrences of a particular word.

    (b) What is prototyping ? Why is it necessary ? Distinguish between the following :

    (i) Scope and visibility of variables

    (ii) Actual and formal arguments.

13. (a) What is a 'slack byte' ? How does it affect the implementation of structures ? Describe three different approaches that can be used to pass structures as function arguments.

    (b) Write a function using pointers to add two matrices and to return the resultant matrix to the calling function.

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